Kumar Miskin

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Summary

Ph.D. candidate in computational materials science (Johns Hopkins) with expertise in machine learning interatomic potentials, Bayesian optimization, active learning, and large-scale atomistic simulation. Five peer-reviewed publications (3 first-author); research spanning a U.S. DOE national laboratory (LLNL), global industry (Sony, TATA Power), and leading academic institutions across three continents. Expecting to graduate Dec 2026; actively seeking full-time ML research and scientific computing roles at technology companies.

Technical Skills

- **ML & Optimization:** Bayesian Optimization, Gaussian Processes, Equivariant Neural Networks, Graph NNs, XGBoost, CNNs, LSTMs, Active Learning, Transfer Learning, Diffusion Models (DDPM), PyTorch, TensorFlow, JAX, Scikit-learn, Genetic Algorithms.
- **Simulation & Modeling:** Molecular Dynamics (LAMMPS), DFT (Quantum ESPRESSO, VASP, Gaussian), Ab Initio MD, NEB, MLIPs (Allegro, MACE, NequIP), Monte Carlo, Force Field Development, ASE, OVITO, VMD.
- **Programming & Data:** Python (Pandas, NumPy, SciPy, Matplotlib, RDKit), C++, MATLAB, SQL (DuckDB), Bash/Linux, Git, LaTeX, HPC/Slurm, Parallel Computing (MPI), Data Analytics & Visualization.
- **Domain Expertise:** Solid-state electrolytes & batteries, halide perovskite semiconductors, defect & radiation physics, shock wave mechanics & spall, materials informatics, computational materials discovery.

Education

- **JOHNS HOPKINS UNIVERSITY, Baltimore, MD** *(Expected: Dec 2026)*
  Ph.D. in Materials Science and Engineering (M.S. obtained 2024)
  Advisor: Prof. Paulette Clancy
- **INDIAN INSTITUTE OF TECHNOLOGY BOMBAY, Mumbai, India** *(Graduated: May, 2022)*
  B.Tech in Metallurgical Eng. & Materials Science + M.Tech in Data Science
  Ranked 1st in Interdisciplinary Dual Degree Program

Research & Work Experience

SMALL-MOLECULE DIFFUSION MODEL | *Equivariant Diffusion for Molecular Conformers* *(Nov 2026)*

- Built a graph-conditioned **DDPM** in **PyTorch** with an E(3)-equivariant noise predictor to generate 3D conformers of small molecules from **RDKit** (ETKDG/MMFF) data; median heavy-atom RMSD 0.76 Å vs. 1.75 Å for random coordinates on 6 held-out molecules (GitHub).

LAWRENCE LIVERMORE NATIONAL LAB | *CCMS Intern* | *Dr. S. Wan & K. Kim* *(Jun 2026 – Present)*

- Training and fine-tuning **Allegro, MACE** and **NequIP** equivariant neural network interatomic potentials on a diverse family of B–H polyhedral anion cluster systems ($NaBH_4$, $LiBH_4$, and complex hydroborate frameworks), enabling large-scale MLIP-MD simulations far beyond DFT timescales.
- Investigating reorientational dynamics of B–H anion clusters and Li^+/Na^+ cation diffusion pathways through hydroborate frameworks via MLIP-MD, establishing structure–transport relationships governing fast-ion conductivity in solid electrolytes.
- Building an end-to-end ML pipeline over MLIP-MD data: feature engineering of structural/dynamical descriptors, feature selection, and **Bayesian Optimization** to efficiently discover novel hydroborate electrolyte compositions with maximized cation conductivity.

JOHNS HOPKINS UNIVERSITY | *Doctoral Researcher* | *Clancy Group* *(Aug 2022 – Present)*

- Architected **PAL 2.0**, a physics-informed active learning system combining **XGBoost**-based feature selection with Bayesian Optimization and Neural Networks, achieving **3x** faster convergence than SMAC/Hyperopt baselines across 12+ benchmarks; published in *Mater. Horiz.* (2023).
- Designed hybrid **GP + Deep Learning** surrogate models for high-dimensional chemical composition spaces, identifying top-performing perovskite photovoltaics while querying **<11%** of the design space via active learning.
- Engineered a parallelized **SQL (DuckDB)** + Python ETL pipeline to process terabyte-scale MD simulation outputs (500k+ atoms), automating defect event detection and cutting analysis time by **90%**.
- Discovered low-energy self-healing defect pathways in $CsPbBr_3$ and radiation response mechanisms in perovskites via DFT/NEB; **>95%** experimental agreement; published as 2 first-author papers (*PCCP* 2025, *J. Mater. Chem. A* 2026).

IIT BOMBAY | *Bachelor's & Master's Thesis* | *Prof. M.P. Gururajan & Prof. Alankar* *(Aug 2021 – May 2022)*

- Built ML classifiers (CNN) to identify NiTi phase transformations from 50+ MD simulations with **90%+** accuracy.
- Modeled the Suzuki segregation effect in FCC Cu using LAMMPS, evaluating stacking fault energies across 50+ impurity configurations via MD–MC simulations and Gibbs free energy analysis.
- **Presented** work at **ICSMA 19** (Int'l Conference on Strength of Materials) in Metz, France.

SONY, Atsugi Technology Center, Japan | *Computational Science Intern* *(Nov 2020 – Dec 2020)*

- Reduced simulation costs by **40%** by building ML surrogate models on 200+ DFT simulations to predict solid-state battery properties (voltage profiles, ion mobility).
- Benchmarked surrogate predictions against held-out DFT calculations across diverse electrolyte chemistries, quantifying model uncertainty to prioritize the most promising candidates for downstream experimental synthesis.

TATA POWER STRATEGIC ENGINEERING DIVISION | *Computer Vision Intern* *(May 2020 – Jul 2020)*

- Deployed a **YOLOv5**-based real-time computer vision system for license plate recognition, improving inference throughput by **35%**.
- Engineered an automated data preprocessing and augmentation pipeline to handle variable lighting, occlusion, and camera angles; improving cross-environment deployment robustness and reducing manual annotation overhead.

1ST ZOOM, Bangalore, India | *Machine Learning Intern* *(May 2020 – Jul 2020)*

- Achieved **94%+** accuracy in audio anomaly detection across 10,000+ samples using MFCC and spectral features; built a real-time signal processing and classification pipeline deployed for production monitoring.
- Conducted ablation studies across MFCC coefficient counts, window sizes, and classifier architectures (SVM, Random Forest, LSTM) to identify the optimal feature–model combination, improving the precision-recall trade-off.

NEW YORK UNIVERSITY | *Summer Research Intern, Prof. Mark Tuckerman* *(Jun 2019 – Jul 2019)*

- Trained regression and classification ML models on X-ray diffraction data from 30+ single crystals to estimate impurity concentrations with **<5%** error; developed feature engineering pipelines over crystallographic structure descriptors.
- Applied PCA and exploratory data analysis to high-dimensional XRD spectra, identifying the most discriminative diffraction features and informing the feature selection strategy that enabled sub-5% prediction error.

Selected Publications

- **K. Miskin**, G. Lemson, T. Arbaugh, E. Ragusa, N.Q. Le, P. Clancy, "How soft metal halide perovskite lattices heal following energetic particle bombardment," *J. Mater. Chem. A*, 2026. [DOI]
- **K. Miskin**, Y. Cao, M. Marland, F. Shaikh, D.T. Moore, J.A. Marohn, P. Clancy, "Low-energy pathways lead to self-healing defects in $CsPbBr_3$," *Phys. Chem. Chem. Phys.*, 2025. [DOI]
- M.S. Priyadarshini, O. Romiluyi, Y. Wang, **K. Miskin**, C. Ganley, P. Clancy, "PAL 2.0: A Physics-Driven Bayesian Optimization Framework for Material Discovery," *Mater. Horiz.*, 2023. [DOI]
- S. Yuan, I. Hsieh **K. Miskin**, K. Kim, L. Wan, "Deciphering the Dynamical and Structural Factors Controlling Na-Ion Transport in Hydroborate-based Solid Electrolytes" *Under Revision*
- **K. Miskin**, M. Zhang, , K. Muly, Joel, K.T. Ramesh, P. Clancy, "How flyer geometry affects shock wave propagation and spall on impact" *In Preparation.*

Achievements & Fellowships

- **Kai Pan & Li Wang Graduate Fellowship**, Johns Hopkins University 2025
- **Donald S. Rodbell Memorial Graduate Fellowship**, Johns Hopkins University 2024, 2025
- **Robert B. Pond Sr. Graduate Fellowship**, Johns Hopkins University 2024
- **Edwin D. & Rachel Lowthian Endowed Fellowship**, Johns Hopkins University 2023
- **Ranked 1st**, Interdisciplinary Dual Degree Program in Materials Science & Machine Learning, IIT Bombay 2022
- **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow**, Dept. of Science & Technology, Govt. of India 2017
- **National Talent Search Scholarship (NTSE)**, NCERT, Govt. of India 2015

Teaching, Mentoring & Service

- **Teaching Assistant**: Molecular Simulations, Phase Transformations, Computational Modeling (3 MSE graduate-level courses, Johns Hopkins University)
- **Research Mentor** – supervised **4+ undergraduate researchers** and **1 graduate researcher** on year-long and summer computational materials projects
- **Lab Manager**, Clancy Research Group – coordinated weekly research meetings and seminars for a multidisciplinary team of 10+ researchers
- **International Student Representative**, Materials Graduate Society, Johns Hopkins University
- **Student Mentor** – Selected as a mentor for IIT Bombay's SMP and DAMP programs, guiding **22 students** through academic planning, campus integration, and personal development while providing ongoing mentorship and support